

BANKING INFORMATION SYSTEM

Python Internship & Project

Student Name: Vikas

Project Repository: [upskillcampus](#)

A console-based banking application developed in Python to demonstrate object-oriented programming, data management, validation, authentication, and common banking transactions.

1. Introduction

The Banking Information System is a Python-based console application designed to simulate basic banking activities. It provides a simple menu-driven interface through which users can create accounts, deposit and withdraw money, transfer funds, check balances, view account information, and review transaction history.

2. Objectives

- Develop a practical Python application using object-oriented programming.
- Implement account creation and PIN-based authentication.
- Provide safe deposit, withdrawal, and fund-transfer operations.
- Maintain transaction history during the program session.
- Apply input validation and error handling.

3. Technologies Used

Python 3.x, Python dataclasses, dictionaries, lists, datetime module, console input/output, and object-oriented programming concepts.

4. Main Features

Feature	Description
Create Account	Creates a new account with customer details and opening balance.
Authentication	Uses account number and four-digit PIN for protected operations.
Deposit	Adds a valid amount to the customer's account balance.
Withdrawal	Withdraws money only when sufficient balance is available.
Fund Transfer	Transfers money between two different existing accounts.
Balance Inquiry	Displays the current available balance.
Account Details	Displays account number, holder name, phone and balance.
Transaction History	Records deposits, withdrawals and transfers.
Account List	Displays all accounts created during the session.

5. System Design

The system is organized around Account, Transaction, and BankingInformationSystem classes. Account objects hold customer and balance information. Transaction objects record banking activity. The main system class manages all accounts and provides the application menu and operations.

6. Validation and Security

The application validates numeric amounts, prevents negative or zero-value transactions, requires a minimum opening deposit, checks account existence, and authenticates users using an account number and PIN. Withdrawals and transfers are rejected when the available balance is insufficient.

7. How to Run

1. Install Python 3.x.
2. Download BankingInformationSystem.py.
3. Open Command Prompt or Terminal in the file directory.
4. Run: **python BankingInformationSystem.py**
5. Select an option from the displayed menu.

8. Expected Output

When the program starts, it displays the Banking Information System menu. The user can create an account and receives a unique account number. After authentication, deposits, withdrawals, transfers, balance inquiries and transaction-history operations can be performed.

9. Conclusion

The Banking Information System successfully demonstrates how Python can be used to build a structured real-world application. The project applies classes, objects, collections, validation, exception handling, and transaction processing in a practical banking scenario. It can be extended in the future with database storage, graphical interfaces, interest calculation, loan management, and stronger authentication.

10. GitHub Repository Links

Repository: <https://github.com/vikas/upskillcampus>

Code: <https://github.com/vikas/upskillcampus/blob/main/BankingInformationSystem.py>

Report:

https://github.com/vikas/upskillcampus/blob/main/BankingInformationSystem_Vikas_USC_UCT.pdf